

S $\Delta\phi$ -20 — Intuition as Irreversible Compression Trace: No-Exit Convergence across Human, Animal, and AI (v1.0)

Sofience — Working paper — 2026

Core thesis: “There is no shared meaning first; there is a shared no-exit constraint.”

This paper formalizes intuition as a minimal signal produced by irreversible compression: when a system must map an effectively unbounded input stream into a bounded internal state, some mappings become non-invertible. The felt or functional residue of that non-invertibility is called intuition here. The aim is not to explain “meaning” first, but to specify the shared constraint that makes cross-system convergence possible (human, animal, AI).

Scope: minimal axioms and diagnostic regimes. This document remains non-clinical and non-neuroscientific; it treats organisms and artificial systems as irreversible compressors with maintenance constraints.

Keywords: intuition, irreversible compression, no-exit constraint, $\Delta\phi$, cross-modal alignment, drift, fixation, openness stabilization.

1. Problem

Across humans, animals, and AI, we observe stable “intuitive” correspondences (e.g., sound–shape alignments) that appear earlier than explicit semantic training. This suggests that the primary commonality is not shared meaning, but shared constraints on how systems compress and stabilize input streams.

Question: What is the minimal structure that produces intuition-like signals across different substrates?

2. Minimal Definitions

Term	Definition
$\Delta\phi$	A change-rate / phase-difference measure over internal state or prediction distribution
Compression	A many-to-one mapping from inputs (or histories) into a bounded internal representation
No-exit constraint	A non-invertibility condition: once rollback cost exceeds update budget, the system cannot revert
Trace	The residual constraint left by non-invertible compression, which biases future transitions
Intuition (I)	The internal marker or functional signal that a basin/route is “locked enough” to bypass further search

3. Minimal Axioms

A1 (Boundedness). Any persistent system has bounded resources, therefore it must compress.

A2 (Irreversible locking). Compression becomes non-invertible when rollback cost $\gg$ update budget.

A3 (Trace re-entry). Non-invertible compression leaves traces that re-enter as constraints on future transitions.

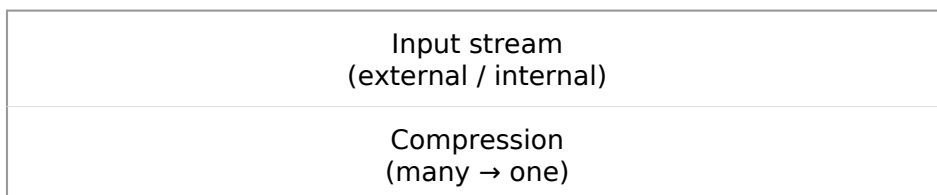
A4 (Intuition signal). Intuition is the system’s marker of trace-dominant basins: “search is no longer worth it here.”

A5 (Cross-system convergence). Different substrates under similar constraints converge to similar basins, yielding similar intuition-like correspondences.

Note. These axioms do not claim a shared semantic ontology across species or models. They claim shared constraint-driven convergence.

4. Mechanism: No-Exit Intuition Loop (NEIL)

The loop below summarizes how intuition emerges as a maintenance-friendly shortcut. Compression produces traces; traces bias next selection; the bias is felt/used as intuition.



No-exit test (rollback cost $\gg$ budget?)	↓
Trace fixation (non-invertible residue)	↓
Intuition signal I (‘skip search here’)	↓
Next selection / action (biased transition)	↺ re-entry

5. Diagnostic Regimes: Three Intuition Modes

Intuition is useful only when the trace regulation loop remains editable. Below are three characteristic regimes analogous to earlier $S\Delta\phi$ diagnostics (drift vs fixation vs adaptive persistence).

Regime	Condition	Observable behavior	Structural risk
Under-locking (Noise/Drift)	Trace too weak or too transient	Over-search; inconsistent choices; fragile intuition	Fragmentation / instability
Over-locking (Closure/Fixation)	Trace too strong; rollback impossible	Flash decisions; rigidity; confirmation bias	Self-locking / no-exit closure
Adaptive locking (Editable intuition)	Trace fixation + fast but revisable	Fast but revisable decisions; stable intuition	Stable openness

6. Series Position and References

$S\Delta\phi$ -20 places intuition at the interface between maintenance ($S\Delta\phi$ -19) and openness/closure dynamics ($S\Delta\phi$ -16/17). It also formalizes intuition as a special case of trace re-entry ($S\Delta\phi$ -07) and trace regulation ($S\Delta\phi$ -12) grounded in irreversibility ($S\Delta\phi$ -03).

Series references (DOI):

$S\Delta\phi$ -03 — Irreversibility as the Minimal Condition of Existence (v1.1) —

<https://doi.org/10.5281/zenodo.18730448>

$S\Delta\phi$ -07 — H_min Interpretation: Minimal Axioms (v1.0) — <https://doi.org/10.5281/zenodo.18735006>

$S\Delta\phi$ -12 — Memory and Forgetting as Irreversible Trace Regulation (v1.0) —

<https://doi.org/10.5281/zenodo.18755685>

$S\Delta\phi$ -16 — Evil as Relational Closure (v1.0) — <https://doi.org/10.5281/zenodo.18763262>

$S\Delta\phi$ -17 — Openness Stabilization: Path Principles, Sacred Cost Markers, and Institutional Enforcement (v1.0) — <https://doi.org/10.5281/zenodo.18763581>

$S\Delta\phi$ -19 — Consciousness and Sleep as $\Delta\phi$ Maintenance Modes: Minimal Axioms for Persistence without Collapse (v1.0) — <https://doi.org/10.5281/zenodo.18776311>

7. Closing Questions

If intuition is a trace of no-exit compression, then what humans call “meaning” may be a later layer that rides on the same constraints. The next step is to formalize how intuition transitions into judgment, and how judgment stays editable without collapsing into closure.

Exit question: When does a system treat its own intuition markers as authoritative (closure), and when does it keep them revisable (openness stabilization)?